SettleIn Project Documentation

Software Engineering Deliverables 1 & 2

DELIVERABLE 1: PROJECT TITLE & PLANNING/REQUIREMENTS ANALYSIS PHASE

1. Project Title

SettleIn: Student Accommodation Marketplace & Booking Reservation System

2. System Description

SettleIn is a web-based booking reservation platform designed to simplify off-campus housing search and reservation for university students. The platform bridges the gap between students seeking verified accommodation and hostel managers/administrators listing available rooms.

- **Front-End (Student Portal):** Enables students to search, filter (by price, location, room type, amenities), view detailed media galleries, reserve rooms, and manage active booking requests.
- **Admin Panel (Management Portal):** Allows hostel administrators and system owners to manage property/room listings, approve or decline student reservation requests, process booking statuses, reset user passwords, and manage user roles.
- **Database:** A centralized relational database storing user accounts, property listings, booking details, payment records, and system audit logs.

3. Group Members & Task Assignments

Member Name	Role / Focus Area	Assigned Tasks
Ian Gachigua (220259)	Integration & QA Lead	Software architecture, RESTful API integration, module contract definitions, system testing, and QA documentation.
Gift Githaka (193923)	Backend & Database Developer	Database schema design (ERD, Normalization), DBMS implementation, server-side business logic, and security.
Yahya Ahmed (220982)	Frontend UI/UX Developer	Student portal UI wireframing/mockups, responsive design, search & filter components, and form validations.

Member Name	Role / Focus Area	Assigned Tasks
Gift Githaka (193923)	Admin Panel Developer	Admin dashboard interface, listing management workflow, reservation processing logic, and report generation.

4. Functional Requirements

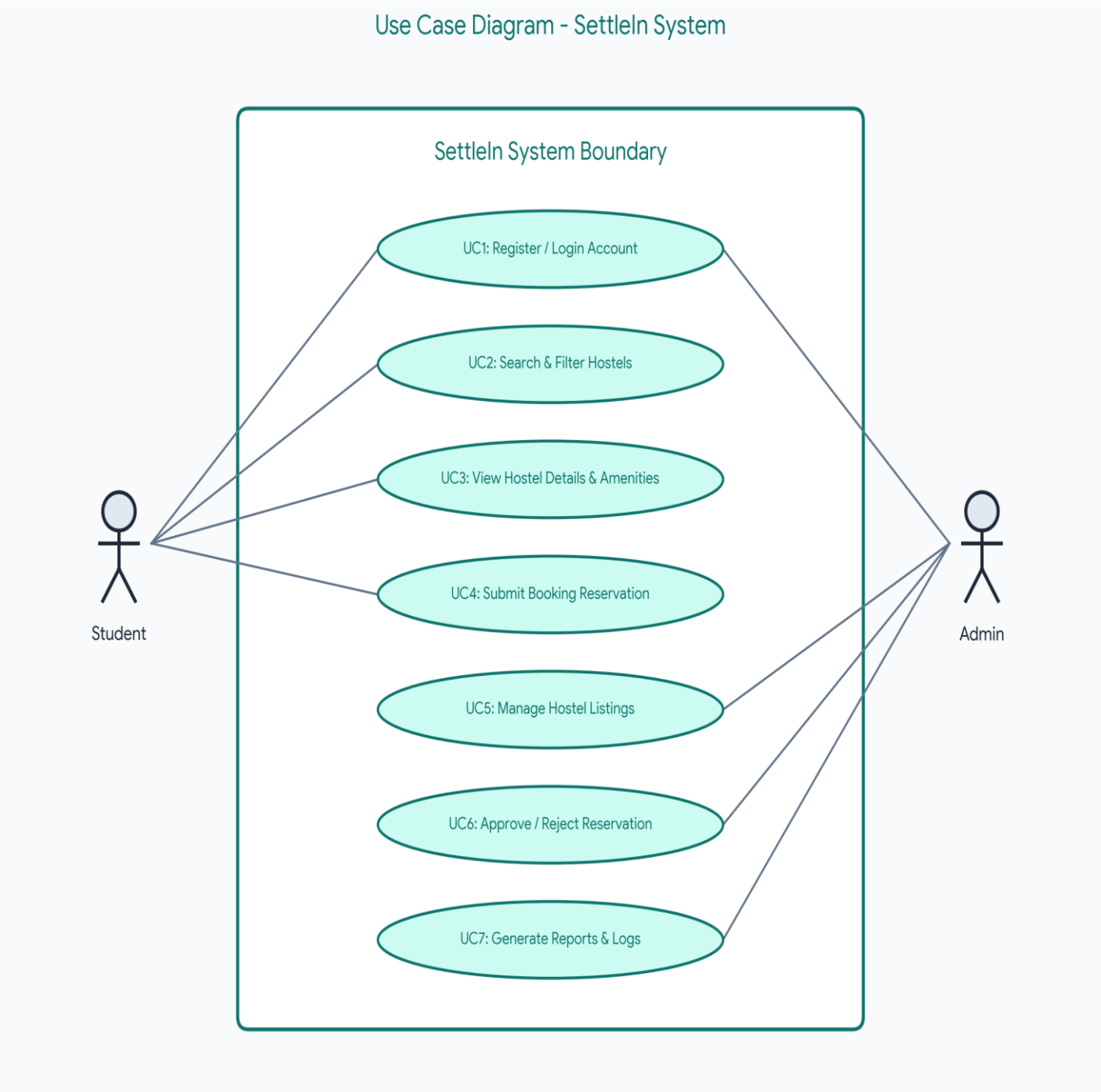
- **1. User Authentication & Profile Management:** Students and Admins must be able to register accounts, log in, log out, update profiles, and request password resets.
- **2. Hostel Search & Filtering:** Students must be able to search and filter available accommodation listings by location, price range, room type (e.g., single, shared), and available amenities.
- **3. Room Reservation & Booking Request:** Students must be able to select an available room, select move-in dates, and submit a booking reservation request.
- **4. Listing & Inventory Management (Admin):** Admins must be able to add, update, view, and delete hostel room listings, image uploads, prices, and vacancy status.
- **5. Reservation Approval & Status Tracking:** Admins must be able to approve, reject, or cancel incoming booking requests, updating the reservation status in real-time for the student.

5. Non-Functional Requirements

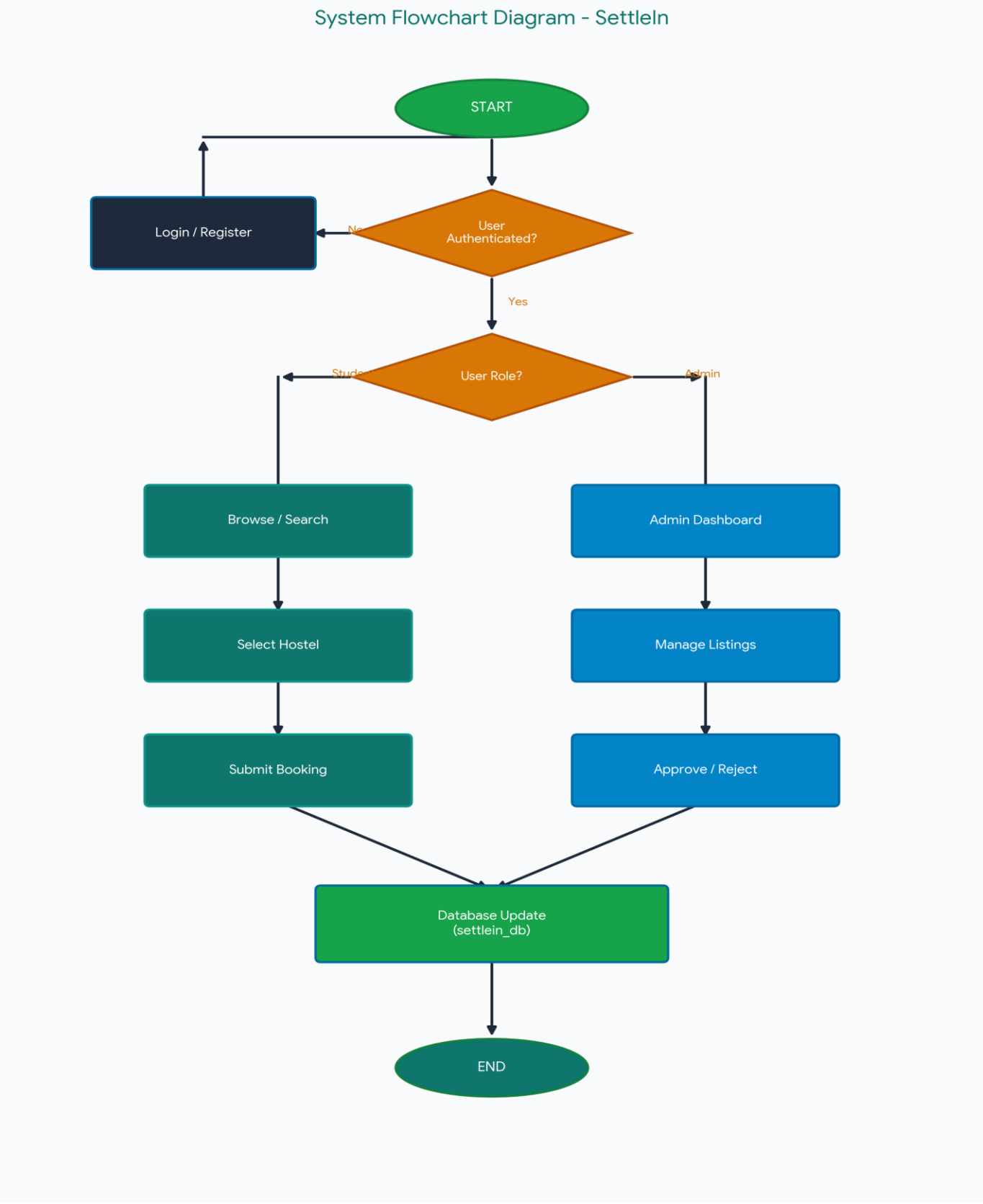
- **1. Security:** All user passwords must be hashed before storage in the database (e.g., using bcrypt), and parameterized SQL/Prepared Statements must be implemented to prevent SQL Injection attacks.
- **2. Performance & Response Time:** Search query results and page load times must execute within 2 seconds under normal user load conditions.
- **3. Usability & Responsive Design:** The front-end user interface must be intuitive, accessible, and fully responsive across mobile devices, tablets, and desktop browsers.

DELIVERABLE 2: SYSTEMS ANALYSIS & SYSTEMS DESIGN PHASES

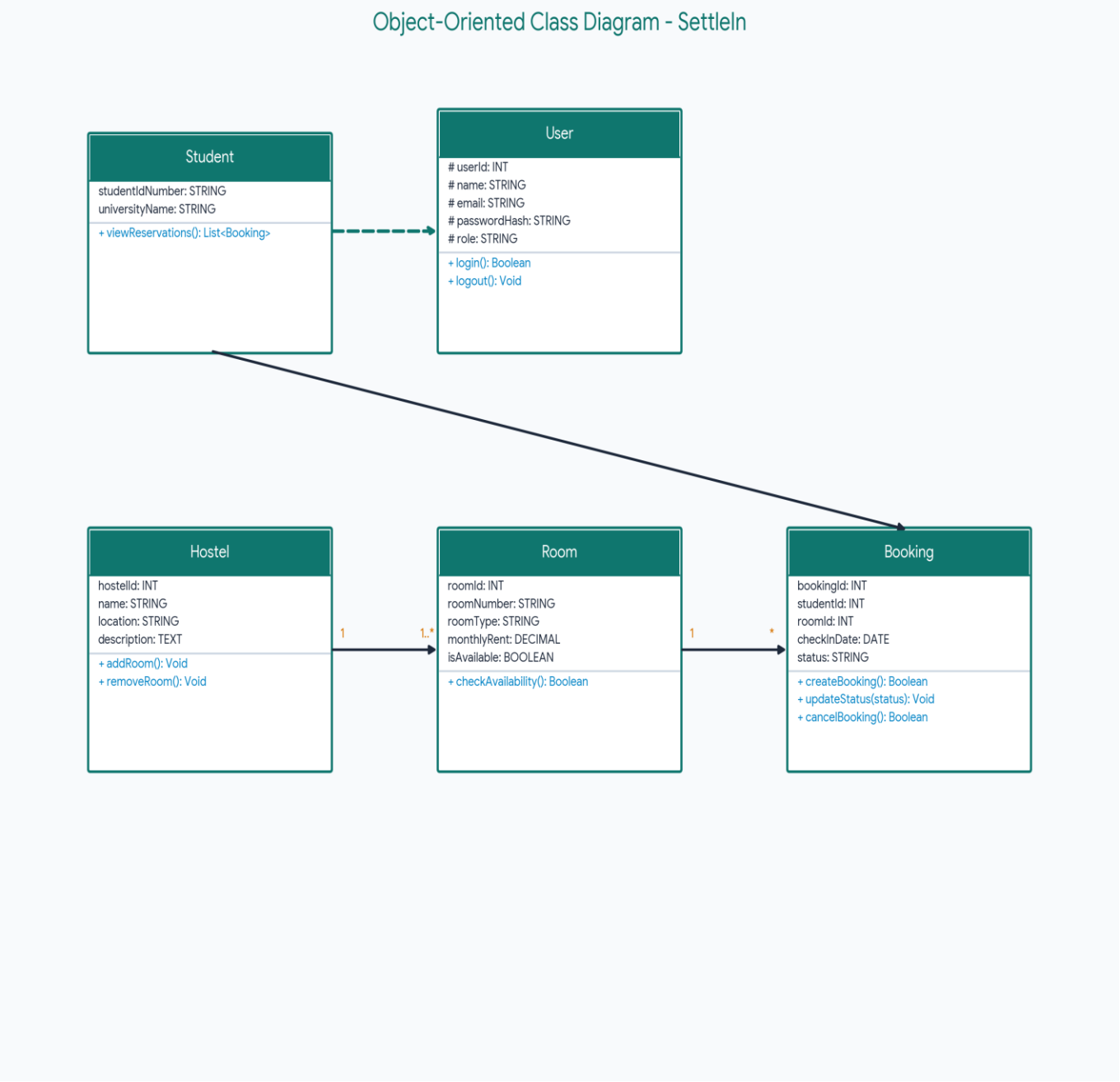
1. Use Case Diagram (Textual & Structural Representation)



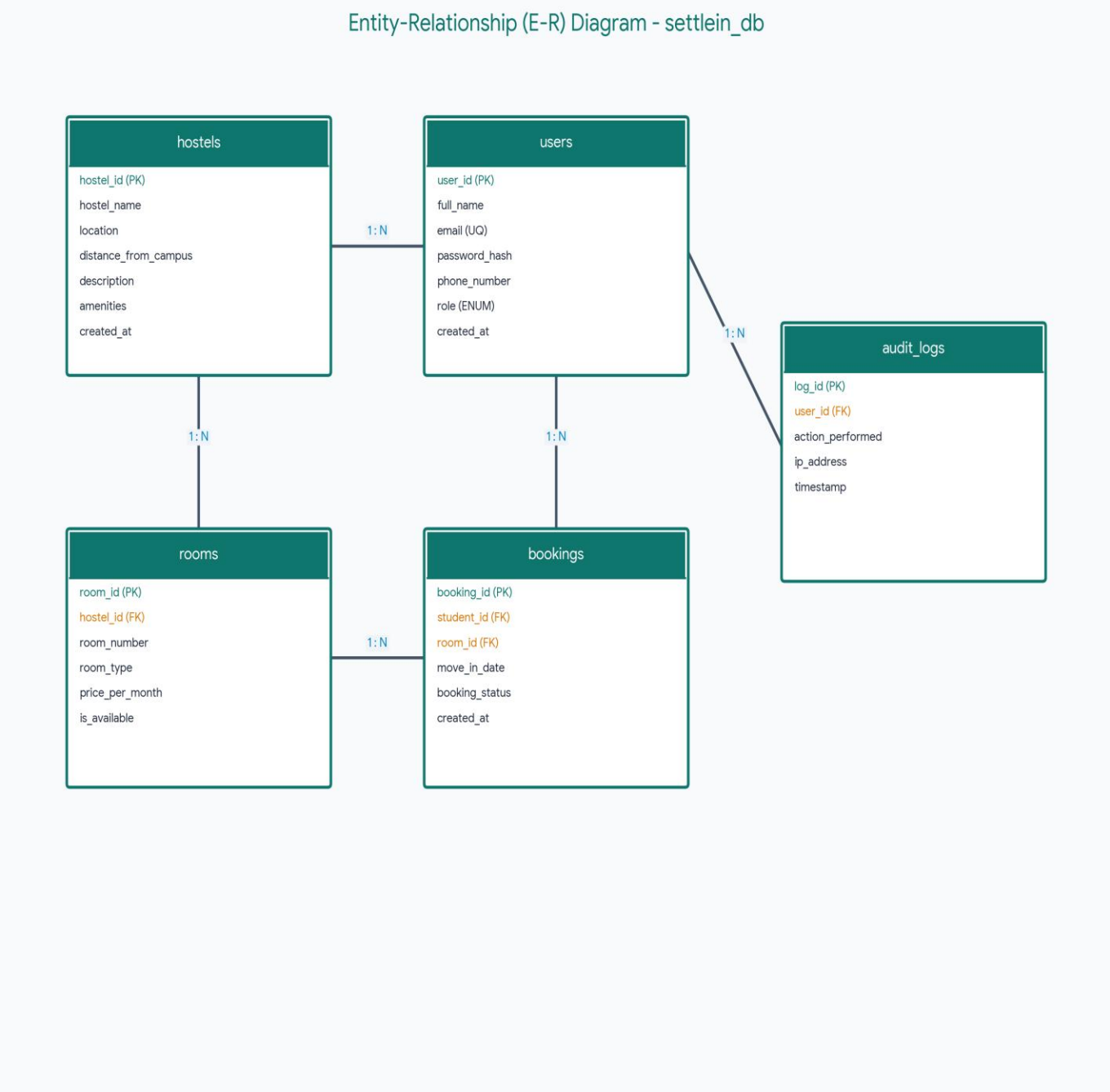
2. System Flowchart Diagram



3. Class Diagram

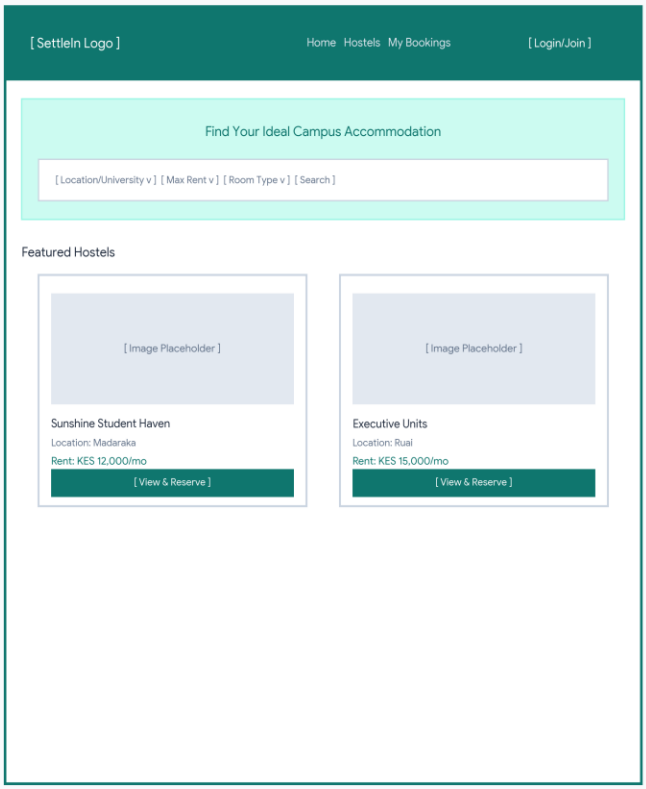


4. Entity-Relationship (E-R) Diagram & Database Schema



5. User Interface Mockups / Wireframe Structure

A. Student Front-End UI Layout



B. Admin Panel Dashboard UI Layout

